

TrES-3b

R-1671

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210331 · created 2026-08-02T10:32:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459304.8573 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.041
Transit depth (Rp/Rs)^2	0.88 +/- 0.78 [%]
Orbital Inclination (inc)	83.14 +/- 2.43
Airmass coefficient 1 (a1)	1.051 +/- 0.016
Airmass coefficient 2 (a2)	-0.0309 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#2 - [242, 110]
Optimal Method	PSF photometry
Transit Duration (day)	0.053 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.094 ± 0.041 vs 0.16309 (deviation 1.43σ)
Duration fit vs published	0.053 d vs 0.0567 d (deviation 0.15σ)
Mid-time O–C	2.2 min
Depth SNR	1.9
Residual scatter	1.5 %

Light curve

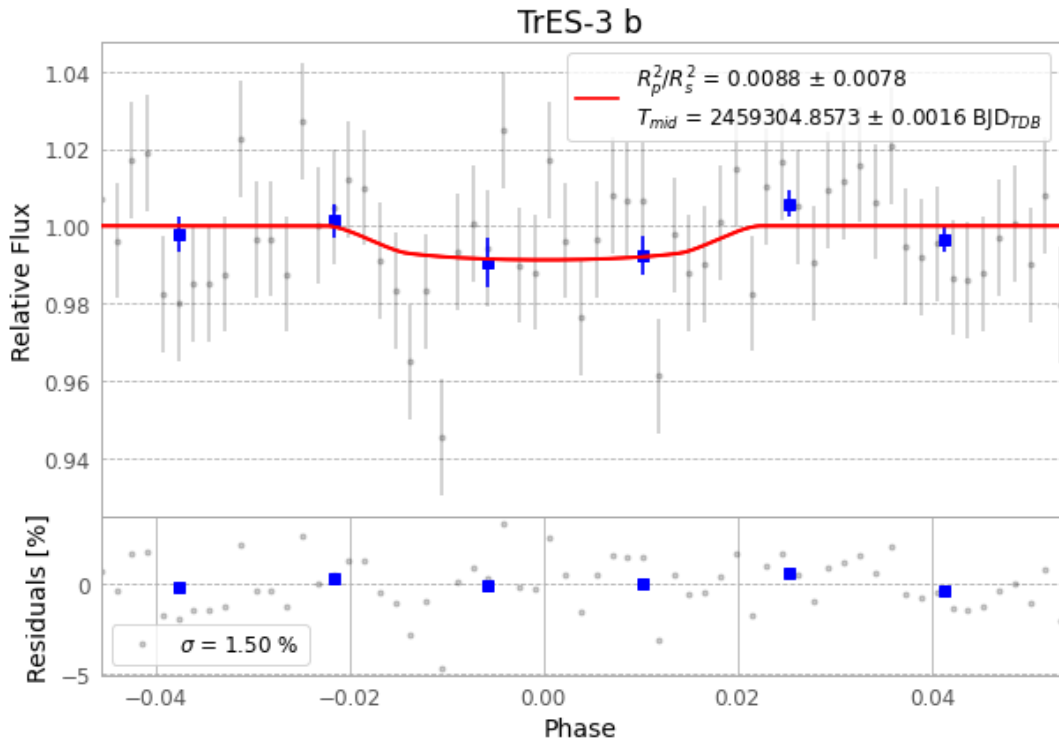


Figure 1 — FinalLightCurve_TrES-3 b_2021-03-31.png (SHA-256 07fa7a837a58ae4c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b9df6637aeee3306...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), TRES-3210331070612.FITS → TRES-3210331101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210331.log (a59737e04133...), FinalLightCurve_TrES-3 b_2021-03-31.png (07fa7a837a58...), FinalLightCurve_TrES-3 b_2021-03-31.csv (1f3bbf90098a...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:32Z.